MQTT Protocol

Release note

Version	Modification	Author	Data
V1.0	First commit	Frank	2024/06/09
V1.1	Add authentication	Frank	2024/08/22
V1.2	Add Instruction Description	Frank	2025/09/03

Instruction Description

Topic	Publish/Subscribe	Description
/TX/{SN}/LOGIN	Publish	Login to the irms
/RX/{SN}/LOGIN	Subscribe	Login result from irms
/TX/{SN}/LOGOUT	Publish	Logout
/TX/{SN}/DEVINFO/HEARTBEAT	Publish	Send heartbeat (interval 3 min)
/TX/{SN}/DEVINFO/BASE	Publish	Send basic infomation
/TX/{SN}/DEVINFO/SIM1	Publish	Send modem1 infomation
/TX/{SN}/DEVINFO/WAN	Publish	Send wan infomation
/TX/{SN}/DEVINFO/LAN	Publish	Send lan infomation
/TX/{SN}/DEVINFO/WLAN	Publish	Send wlan infomation
/TX/{SN}/DEVINFO/NET_QUALITY	Publish	Send network quality(delay/lost/jitter)
/TX/{SN}/DEVINFO/GPS	Publish	Send GPS infomation
/RX/{SN}/EVENT/GETFILE	Subscribe	Request config file from irms
/TX/{SN}/EVENT/GETFILE	Publish	Send currect config file
/RX/{SN}/EVENT/SETFILE	Subscribe	Set config file from irms
/TX/{SN}/EVENT/SETFILE	Publish	Send config set result to irms
/TX/{SN}/INDUSTRY/GET	Publish	Send gpio/serial infomation
/RX/{SN}/INDUSTRY/GET	Subscribe	Request gpio/serial from irms
/RX/{SN}/INDUSTRY/SET	Subscribe	Update gpio/serial config from irms

Topic	Publish/Subscribe	Description
/TX/{SN}/INDUSTRY/SET	Publish	Send gpio/serial config set result
/TX/{SN}/EVENT/LOGINFO	Publish	Send web/cli login event
/RX/{SN}/EVENT/SHELL	Subscribe	Reboot from irms
/TX/{SN}/EVENT/SHELL	Publish	Send reboot result
/RX/{SN}/UPGRADE	Subscribe	Upgrade from irms
/TX/{SN}/UPGRADE	Publish	Send upgrade result to irms
/TX/{SN}/ALARM	Publish	Send alarm to irms
/RX/{SN}/ALARM	Subscribe	Alarm result from irms

MQTT Authentication

Username&password

Parameter	Format	Description
username	string	irmsnetwork
password	string	9764a18f38ecfb420f129be10f5de41f

Certificate

```
-----BEGIN CERTIFICATE-----
MIIDozCCAougAwIBAgIJANg/FfmIMcOfMA0GCSqGSIb3DQEBCwUAMGgxCzAJBgNV
BAYTAkNOMQ4wDAYDVQQIDAVIdWJlaTEOMAwGA1UEBwwFV3VoYW4xETAPBgNVBAoM
CEhvbmdkaWFuMQ4wDAYDVQQLDAVjbG91ZDEWMBQGA1UEAwwNMTcyLjE2LjIyLjE1
MDAeFw0xOTExMDYwNjM2NTlaFw0yOTExMDMwNjM2NTlaMGgxCzAJBgNVBAYTAkNO
MQ4wDAYDVQQIDAVIdWJlaTEOMAwGA1UEBwwFV3VoYW4xETAPBgNVBAoMCEhvbmdk
aWFuMQ4wDAYDVQQLDAVjbG91ZDEWMBQGA1UEAwwNMTcyLjE2LjIyLjE1MDCCASIw
DQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBAJij6dE2tlJzZnHGyXL6pgDOU6dJ
q3jXcQxR/ct6DLNy57OCsh/QihHUunoQmtRa3qfGHDoAyTagMX55CMmOGl72+MM1
JyzqQyM74wHPrMXsYus5u3yV0DJbdpMlS6MUez6Kpm+aVoPPdYiFRUHuvyRGtyf+
9UItliCducufgzQoJar6rvWK8ngqDwaDFKRfIXMkbsJSDgH+5qVt4uh0RkortA4P
BN398x5Ftzqi/nBes7cqhCRzx/fw1ssIfjQUCta0qcSA/BCWfLgZL3D0u3CJEo3C
k6UO3C8t9UE7e7DF5Aqs34R7MscsqjLr6bYWV4ufZRs6z5UACe5y5gHTfU8CAwEA
AaNQME4wHQYDVR0OBBYEFIt9v20b1U7ssX7f2FM8f+6zNUSLMB8GA1UdIwQYMBaA
FIt9v20b1U7ssX7f2FM8f+6zNUSLMAwGA1UdEwQFMAMBAf8wDQYJKoZIhvcNAQEL
BQADggEBADilwgTlcyK4we/6HAmOgIpIZGDW8JkQItH5thKYJ7aQaeJVnCWh8DlL
kXeHID9YZrhMOCq28LLBYo41dH5pBRbIg97IhZIOzMufWRwUqaAM/BeMfwTw/Ucr
6RIMowXrWeDdmDn4TVTtFOL9t8DPr3QsTEJI7b2lnWf5MRbmszZIwJCQqxdFhckQ
PZ+PutwjPLBvhSKtC1XqYWqVCof5sGHlrnvb6UGc8q/HuBlOllOQNbZGX1bK3snR
d61OZtImha/JsTORvzpOkZOt0z9NH45Qf7tiBHW9uPH5GK7zutWmHsvppJdKdiKN
PLFPzR6lM8PgLmFZdjq3rJYpVPrfRCI=
-----END CERTIFICATE-----
```

mosquitto auth code

```
int mqtt_auth_set(struct mosquitto *mosq)
{
    /* username&password set */
    mosquitto_username_pw_set(mosq, "irmsnetwork",
"9764a18f38ecfb420f129be10f5de41f");

    /* certificate set */
    mosquitto_tls_opts_set(mosq, 0, "tlsv1.2", NULL);
    /* save Certificate content to /bus/rootCA.crt */
    mosquitto_tls_set(mosq, "/bus/rootCA.crt", NULL, NULL, NULL, NULL);
    mosquitto_tls_insecure_set(mosq, true);
}
```

Login

Topic	/{TX|RX}/{SN}/LOGIN

Parameter	Format	Description
login	string	timestamp

/TX/{SN}/LOGIN

> **Note**
>
> The content is empty, just send topic

```

```

/RX/{SN}/LOGIN

```
{
    "login":"1580381938"
}
```

Logout

Topic	/{TX|RX}/{SN}/LOGOUT

/TX/{SN}/LOGOUT

> **Note**
>
> The content is empty, just send topic

```

```

DEVINFO

HEARTBEAT

Topic	/{TX|RX}/{SN}/DEVINFO/HEARTBEAT

Parameter	Format	Description
power_time	number	device uptime, format is second

/TX/{SN}/DEVINFO/HEARTBEAT

```
{
    "power_time": 3625
}
```

BASE

Topic	/{TX|RX}/{SN}/DEVINFO/BASE

Parameter	Format	Description
devsn	string	device serial number
softver	string	soft version
hardver	string	hard version
mem_use	string	memory use percentage
cpu_use	string	cpu use percentage
longitude	string	longitude
latitude	string	latitude
nettype	string	WAN|WLAN|5G|4G|3G|2G
cursim	string	current sim card, sim1/sim2
failover-current	string	
failover-primary	string	
failover-backup	string	

/RX/{SN}/DEVINFO/BASE

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/BASE

```
{
        "devsn": "RX400Q1909210009",
        "softver":"V2.2.1",
        "hardver":"V13",
        "mem_use":"12%",
        "cpu_use":"30%",
        "longitude":"113.88308",
        "latitude":"22.553292",
        "nettype":"4G",
        "cursim":"sim1",
        "failover-current":"eth0",
        "failover-primary":"eth0",
        "failover-backup":"modem"
}
```


SIM{1|2}

Topic	/{TX|RX}/{SN}/DEVINFO/SIM{1|2}

Parameter	Format	Description
csq	string	signal csq
nettype	string	network type
lac	string	
ci	string	
sid	string	
bid	string	
nid	string	
ip	string	
iccid	string	
imsi	string	
imei	string	
netband	string	
rsrp	string	
rsrq	string	
sinr	string	
flowtx	string	This is time period traffic,unit is KB
flowrx	string	This is time period traffic,unit is KB
operator	string	

Parameter	Format	Description
uptimes	string	
offtime	string	
module_version	string	
snr	string	

/RX/{SN}/DEVINFO/SIM1

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/SIM1

```
{
    "csq":"31",
    "nettype":"4G",
    "lac":"9537",
    "ci":"112981003",
    "sid":"",
    "bid":"",
    "nid":"",
    "ip":"10.239.53.79",
    "iccid":"89860117851125692043",
    "imsi":"460010388623133",
    "imei":"867377023567328",
    "netband":"band3",
    "rsrp":"-94",
    "rsrq":"-5",
    "sinr":"3",
    "flowtx":"",
    "flowrx":"",
    "operator":"CHN-UNICOM",
    "uptimes":"2",
    "offtime":"135",
    "module_version":"EC25EFAR02A08M4G",
    "snr":"1"
}
```

WAN

Topic	/{TX|RX}/{SN}/DEVINFO/WAN

Parameter	Format	Description
status	string	
mode	string	

Parameter	Format	Description
ip	string	
flowtx	string	This is time period traffic,unit is KB
flowrx	string	This is time period traffic,unit is KB
wan1	string	up|down

/RX/{SN}/DEVINFO/WAN

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/WAN

```
{
    "status":"1",
    "mode":"static",
    "ip":"192.168.10.1",
    "flowtx":"100",
    "flowrx":"200",
    "wan1":"up/100Mbps/full-duplex",
}
```

LAN

Topic	/{TX|RX}/{SN}/DEVINFO/LAN

Parameter	Format	Description
status	string	
ip	string	
lan1	string	up|down
lan2	string	up|down
lan3	string	up|down
lan4	string	up|down

/RX/{SN}/DEVINFO/LAN

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/LAN

```
{
    "status":"1",
    "ip":"192.168.8.1",
    "lan1":"up/1000Mbps/full-duplex",
    "lan2":"down",
    "lan3":"down",
    "lan4":"down"
}
```

WLAN

Topic	/{TX|RX}/{SN}/DEVINFO/WLAN

Parameter	Format	Description
status	string	
ip	string	
channel	string	
number	string	user number

/RX/{SN}/DEVINFO/WLAN

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/WLAN

```
{
    "mode":"ap",
    "ip":"192.168.8.191",
    "channel":"1",
    "number":"1"
}
```

NET_QUALITY

Topic	/{TX|RX}/{SN}/DEVINFO/NET_QUALITY

Parameter	Format	Description
delay	string	
loss_rate	string	
jitter	string	

/RX/{SN}/DEVINFO/NET_QUALITY

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/NET_QUALITY

```
{
    "delay":"23ms",
    "loss_rate":"0%",
    "jitter":"10ms"
}
```

CLIENT_LIST

Topic	/{TX|RX}/{SN}/DEVINFO/CLIENT_LIST

Parameter	Format	Description
ip	string	
mac	string	
tport	array	tcp port array
uport	array	udp port array

/RX/{SN}/DEVINFO/CLIENT_LIST

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/CLIENT_LIST

```
{
    "client":[
        {"ip":"192.168.8.2", "mac":"40:8d:5c:e0:e2:24", "tport":[22,23,80],
"uport":[]},
        {"ip":"192.168.8.3", "mac":"40:8d:5c:e0:e2:22", "tport":[22,23,80],
"uport":[]}
    ]
}
```

GPS

Topic	/{TX|RX}/{SN}/DEVINFO/GPS

Parameter	Format	Description
fix_status	string	GPS fix time

Parameter	Format	Description
used	number	statellites used
visible	number	statellites visible
lat	float	latitude
lng	float	longitude
height	float	Height
speed	float	Speed
direction	float	Direction

/RX/{SN}/DEVINFO/GPS

```
{
    "get":""
}
```

/TX/{SN}/DEVINFO/GPS

> **Note**
>
> Upload every minute

```
{
    "fix":"12:32:00",
    "used":6,
    "visible":8,
    "lat": -34.1132,
    "lng": 151.4283,
    "height": 0.00,
    "speed": 0.0,
    "direction": 0.0
}
```

CONFIG

Set config

Topic	/{TX|RX}/{SN}/CONFIG

Parameter	Format	Description
cli	string	format is hex string

/RX/{SN}/CONFIG

> **Note**

> we can convert string and hex by below url
>
> https://codebeautify.org/hex-string-converter
>
> https://codebeautify.org/string-hex-converter

```
{
    "cli":"736572766963656d326d7065657231322e31322e31..."
}
```

/TX/{SN}/CONFIG

```
{
    "cli":"succeed"
}
```

Upload config

Topic	/{TX|RX}/{SN}/CONFIG

Parameter	Format	Description
diff	string	format is hex string

/TX/{SN}/CONFIG

> **Note**
>
> when config change, upload the diff with last config

```
{
    "diff":"2d2d2d636c692e636f6e66323032302d30312..."
}
```

/RX/{SN}/CONFIG

```
{
    "diff":"succeed"
}
```

INDUSTRY

GET

Topic	/{TX|RX}/{SN}/INDUSTRY/GET

Parameter	Format	Description
gpio.direction	string	out|in
gpio.alias	string	name

Parameter	Format	Description
gpio.value	number	1|0
gpio.ware	string	100,-100
serial.name	string	name
serial.status	number	1|0
serial.type	string	RS232|RS485
serial.rate	number	300|600|1200...
serial.parity	string	none|even|odd
serial.databits	number	5|6|7|8
serial.stopbits	number	1|2

/RX/{SN}/INDUSTRY/GET

```
{
    "get":["gpio","serial"]
}
```

/TX/{SN}/INDUSTRY/GET

> **Note**
>
> Upload gpio status every hour or value change

```
{
    "gpio":{
        "do1":{
            "direction":"out",
            "alias":"do1",
            "value":1
        },
        "di1":{
            "direction":"in",
            "alias":"di1",
            "value":1
        }
    },
    "serial":{
        "serial1":{
            "name":"serial1",
            "status":1,
            "type":"RS232",
            "rate":9600,
            "parity":"none",
            "databits":8,
            "stopbits":1
        }
    }
```

```
}
```

SET

Topic	/{TX|RX}/{SN}/INDUSTRY/SET

/RX/{SN}/INDUSTRY/SET

> **Note**
>
> DI can't set

```
{
    "gpio":{
        "do1":{
            "alias":"test1",
            "value":1
        },
        "do2":{
            "alias":"test1",
            "wave":"100,-200,100,-800"
        },
        "di1":{
            "alias":"test2"
        }
    },
    "serial":{
        "serial1":{
            "name":"serial1",
            "status":1,
            "type":"RS232",
            "rate":9600,
            "parity":"none",
            "databits":8,
            "stopbits":1
        }
    }
}
```

/TX/{SN}/INDUSTRY/SET

```
{
    "gpio":"succeed",
    "serial":"succeed"
}
```

UPLOAD

Topic	/{TX|RX}/{SN}/INDUSTRY/UPLOAD

/TX/{SN}/INDUSTRY/UPLOAD

```
{
    "gpio":{
        "do1":{
            "direction":"out",
            "alias":"test1",
            "value":1
        },
        "di1":{
            "direction":"in",
            "alias":"test2",
            "value":1
        }
    }
}
```

EVENT

LOGINFO

Topic	/{TX|RX}/{SN}/EVENT/LOGINFO

Parameter	Format	Description
username	string	
mode	string	local|radius|tacacs+
authority	string	admin|guest
name	string	
ip	string	
mac	string	
time	string	timestamp
methods	string	web|cli
result	string	succeed|fail

/TX/{SN}/EVENT/LOGINFO

> **Note**
>
> when login by web/cli, upload login info

```
{
    "username":"admin",
    "mode":"local",
    "authority":"admin",
    "name":"pc-13",
    "ip":"192.168.8.2",
    "mac":"20:15:01:01:00:00",
    "time":"1517486527",
    "methods":"web",
    "result":"succeed"
 }
```

/RX/{SN}/EVENT/LOGINFO

```
{
    "time":"1517486527"
}
```

GETFILE

Topic	/{TX|RX}/{SN}/EVENT/GETFILE

Parameter	Format	Description
filename	string	
data	string	16bit hex

/RX/{SN}/EVENT/GETFILE

> **Note**
>
> RMS get device config by this api, so when you receivce GETFILE with '/etc/rxconfig/cli.conf', please upload your config file

```
{
    "filename":"/etc/rxconfig/cli.conf"
}
```

/TX/{SN}/EVENT/GETFILE

> **Note**
>
> please convert string to hex, https://codebeautify.org/string-hex-converter

```
{
    "filename":"/etc/rxconfig/cli.conf",
    "data":"16bit hex"
}
```

SETFILE

Topic	/{TX|RX}/{SN}/EVENT/SETFILE

Parameter	Format	Description
filename	string	
data	string	16bit hex
result	string	succeed|fail

/RX/{SN}/EVENT/GETFILE

> **Note**
>
> RMS set device config by this api, so when you receivce SETFILE with '/etc/rxconfig/cli.conf', please save the file to your config

```
{
    "filename":"/etc/rxconfig/cli.conf",
    "data":"16bit hex"
}
```

/TX/{SN}/EVENT/GETFILE

```
{
    "result":"succeed"
}
```

SHELL

Topic	/{TX|RX}/{SN}/EVENT/SHELL

Parameter	Format	Description
shell	string	reboot|kill|disable|sed
time	string	timestamp

/RX/{SN}/EVENT/GETFILE

```
{
    "shell":"reboot",
    "time":"1581327480",
}
```

/TX/{SN}/EVENT/GETFILE

```
{
    "shell":"succeed"
    "time":"1581327480",
}
```

GETPATCH

Topic	/{TX|RX}/{SN}/EVENT/GETPATCH

Parameter	Format	Description
getpatch	NULL	
avilable	string	Remaining space(KB)

/RX/{SN}/EVENT/GETPATCH

```
{
    "getpatch":""
}
```

/TX/{SN}/EVENT/GETPATCH

```
{
    "avilable":"3837952",
    "ipsec_init":"v1.0.0",
    "modem":"v1.0.2"
}
```

GPSTRACKING

Topic	/{TX|RX}/{SN}/GPSTRACKING

Parameter	Format	Description
id	number	1-65535
timestamp	number	timestamp
accuracy	number	Displacement distance(metre)

/TX/{SN}/EVENT/GPSTRACKING

```
{
    "id": 4,
    "timestamp": 1704102962,
    "accuracy":24
    "gps": {
        "lat": -34.1132,
        "lng": 151.4283,
        "height": 0.00,
```

```
        "speed": 0.0,
        "direction": 0.0
    },
    "net":{
        "sim": "sim1",
        "nettype": "LTE",
        "csq": 26,
        "band": "Band 1",
        "cell": 20593934,
        "lac": 12012,
        "plmn": 46001,
        "carrier": "CHN-TELECOM",
        "rsrp": -82,
        "rsrq": -10,
        "sinr":11,
        "rssi":1,
        "ecio":1
    }
}
```

/RX/{SN}/EVENT/GPSTRACKING

```
{
    "id":4
}
```

UPGRADE

Topic	/{TX|RX}/{SN}/UPGRADE

Parameter	Format	Description
patch/firmware/boot/openvpn	string	upgrade type
configure	string	upload config

/RX/{SN}/EVENT/UPGRADE

> **Note**
>
> Upgrade by this api, download by curl or wget, and url will use http or https

```
{
    "firmware":"https://xxx/xxx/xxx.bin"
}
```

/TX/{SN}/EVENT/UPGRADE

```
{
    "result":"succeed"
}
```

ALARM

Topic	/{TX|RX}/{SN}/ALARM

Parameter	Format	Description
id	number	1-65535
time	string	timestamp
event.type	string	
event.object	string	
event.action	string	
event.data	string	

network

object	action
sim1-status	up+ip|down
sim2-status	up+ip|down
net-switch	modem->eth0
sim-switch	sim1->sim2
sim-redial	192.168.8.2->192.168.8.3
linkbackup-switch	eth0->modem

vpn

object	action
vpdn-xxx	up+ip|down
tunnel-xxx	up+ip|down
ipsec-xxx	up|down
openvpn	up+ip|down

system

object	action
power-on	
power-off	
factory	cli|button|gui

object	action
lan(1-4)-status	up|down
wan(1-4)-status	up+ip+dhcp|down
geo-fencing	116.366,39.920->116.266,39.620(23 meters)
usb-status	up|down

/TX/{SN}/ALARM

```
{
    "id":1,
    "time":"1624370557",
    "event":{
        "type":"vpn",
        "object":"ipsec-1",
        "action":"up",
        "data":""
    }
}
```

/RX/{SN}/ALARM

> **Note**
>
> Must receive same id then we can delete the alarm id

```
{
    "id":1
}
```

PROXY

Socket

Topic	/{TX|RX}/{SN}/PROXY/IP/DPORT/SPORT

/RX/{SN}/PROXY/127.0.0.1/80/42512

```
raw data
```

/TX/{SN}/PROXY/127.0.0.1/80/42512

```
raw data
```

Serial

> **Note**
>
> Need config by protocol INDUSTRY first

Topic	/{TX|RX}/{SN}/PROXY/serial{i}

/RX/{SN}/PROXY/serial1

```
Hex data
```

/TX/{SN}/PROXY/serial1

```
Hex data
```

SDWAN

Topic	/{TX|RX}/{SN}/SDWAN/VPNSTATUS

Parameter	Format	Description
type	string	openvpn|ipsec|pptp|...
status	number	1|0
ip	string	
mac	string	

/TX/{SN}/SDWAN/VPNSTATUS

```
{
    "type":"openvpn",
    "status":"1",
    "ip":"192.168.120.6",
    "mac":"40:8d:5c:e0:e2:12"
}
```